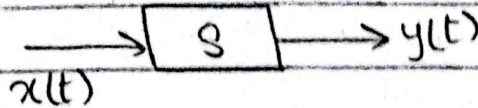
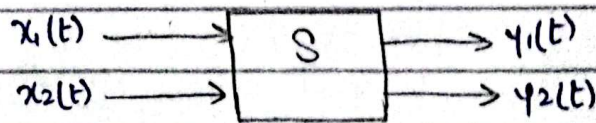


First Week

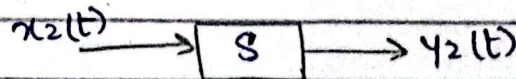
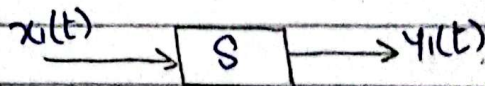
System



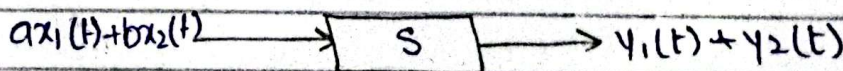
- Single input single output (SISO)
- Multiple input multiple output (MIMO)



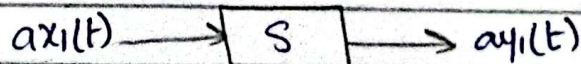
LINEAR System:



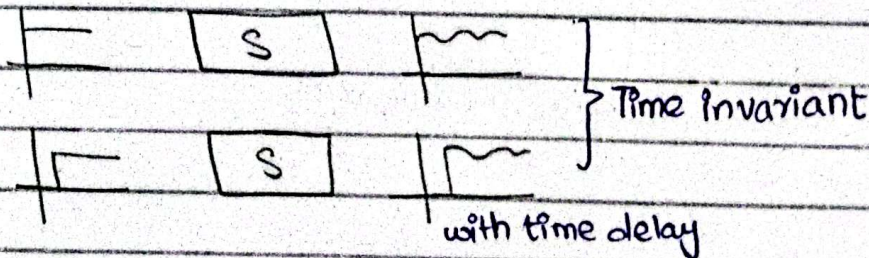
Properties:



↳ addition



↳ homogeneity



* no feedback system — open loop

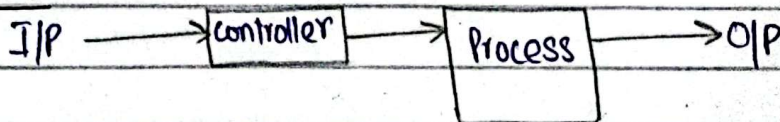
Components of a Control System:

- Input
- Output
- ways to achieve input & output
- control action

Basic Technologies

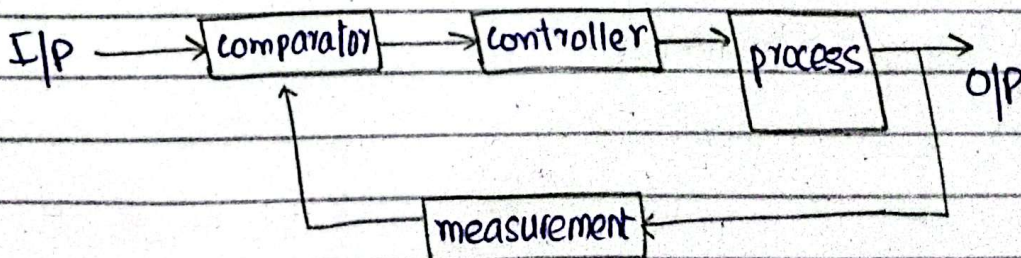
- Plant / Process: portion of the system which is to be controlled or regulated.
- Controller: controls the plant / process
- Input
- output
- disturbance

Input:

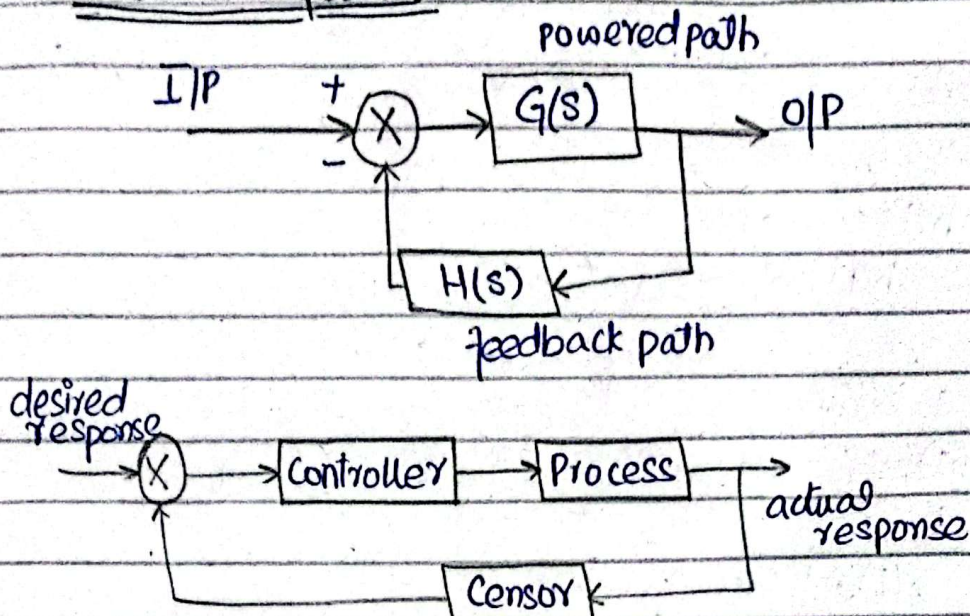


Open loop: O/P is neither measured nor feedback.

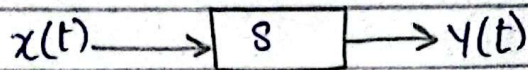
Closed Loop Control System



Feedback System:



Chapter no 2



Transfer function: $T.F = \frac{Y(s)}{X(s)}$

Laplace transform: (covers a broader range of system)

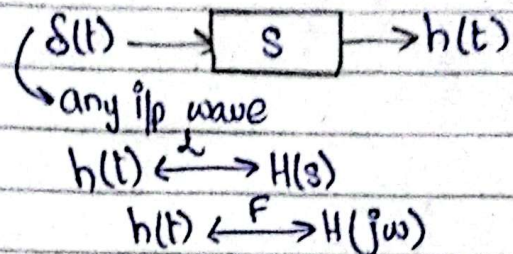
$$X(s) = \int_{-\infty}^{\infty} x(t)e^{-st} dt$$

Fourier transform: (for signal analysis)

$$X(j\omega) = \int_{-\infty}^{\infty} x(t)e^{-j\omega t} dt$$

* Fourier is a subset of Laplace transform

Review of Laplace Transform



$\Rightarrow$ if $s = \sigma + j\omega$

$$X(s) = \int_{-\infty}^{\infty} x(t) e^{-st} dt = \int_{-\infty}^{\infty} x(t) e^{-\sigma t} e^{-j\omega t} dt$$

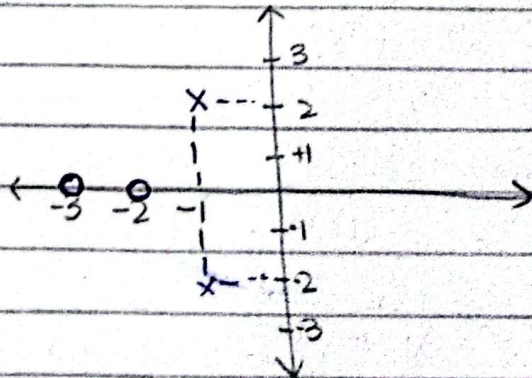
$$= \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt \quad (\text{Fourier})$$

Example 1 $H(s) = \frac{(s+2)(s+3)}{(s+1+2j)(s+1-2j)}$

zeros $\Rightarrow s = -2, s = -3$; poles $\Rightarrow s = -1-2j, s = -1+2j$

zeros — represented by 'o'

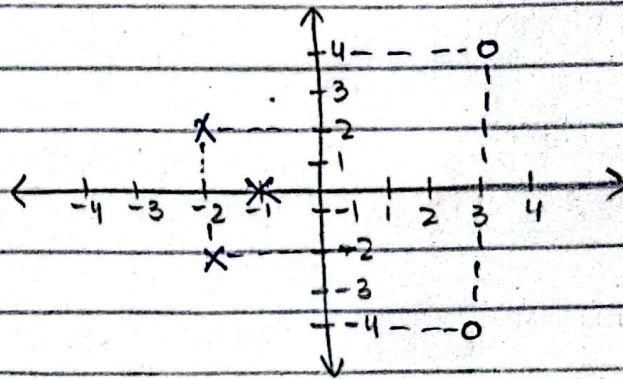
poles — represented by 'x'



Eg #2

$$H_1(s) = \frac{(s-3+4j)(s-3-4j)}{(s+1)(s+2+2j)(s+2-2j)}$$

$$\begin{array}{llll} s-3+4j=0 & ; & s-3-4j=0 & ; & s=-1 & ; & s+2+2j=0, & s+2-2j=0 \\ s=3-4j & & s=3+4j & & & & s=-2-2j & s=-2+2j \end{array}$$



Impulse related properties

1) $x_1(t) = \delta(t)$

$$X_1(s) = \int_{-\infty}^{\infty} \delta(t) e^{-st} dt = e^{-s(0)} = 1$$

2) $x_2(t) = u(t)$

$$X_2(s) = \int_{-\infty}^{\infty} u(t) e^{-st} dt$$

$$= \int_0^{\infty} e^{-st} dt = \left. \frac{e^{-st}}{-s} \right|_0^{\infty} = -\frac{1}{s} (e^{-\infty} - e^0) = \frac{1}{s}$$

3) $x_3(t) = t u(t)$

$$X_3(s) = \int_{-\infty}^{\infty} t u(t) e^{-st} dt = \int_0^{\infty} t e^{-st} dt$$

$$= \left. \frac{t e^{-st}}{-s} \right|_0^{\infty} - \int_0^{\infty} \frac{1 \cdot e^{-st}}{-s} dt = \frac{1}{s} \left. \frac{e^{-st}}{-s} \right|_0^{\infty} = \frac{1}{s^2}$$

$$4) x_4(t) = e^{-at} u(t)$$

$$\begin{aligned} X_4(s) &= \int_{-\infty}^{\infty} e^{-at} u(t) e^{-st} dt \\ &= \int_0^{\infty} e^{-(s+a)t} dt = \left. \frac{e^{-(s+a)t}}{-(s+a)} \right|_0^{\infty} = \frac{1}{s+a} \end{aligned}$$

$$5) x_5(t) = \sin \omega t u(t)$$

$$\therefore \sin \omega t = \frac{e^{j\omega t} - e^{-j\omega t}}{2j}$$

$$= \frac{e^{j\omega t} - e^{-j\omega t}}{2j} u(t)$$

$$\therefore \cos \omega t = \frac{e^{j\omega t} + e^{-j\omega t}}{2}$$

$$X_5(s) = \frac{1}{2j} \int_0^{\infty} e^{j\omega t} e^{-st} dt - \frac{1}{2j} \int_0^{\infty} e^{-j\omega t} e^{-st} dt$$

$$= \frac{1}{2j} \int_0^{\infty} e^{-(s-j\omega)t} dt - \frac{1}{2j} \int_0^{\infty} e^{-(s+j\omega)t} dt$$

$$= \frac{1}{2j} \left[\left. \frac{e^{-(s-j\omega)t}}{-(s-j\omega)} \right|_0^{\infty} - \left. \frac{e^{-(s+j\omega)t}}{-(s+j\omega)} \right|_0^{\infty} \right]$$

$$X_5(s) = \frac{1}{2j} \left[\frac{1}{s-j\omega} - \frac{1}{s+j\omega} \right] = \frac{1}{2j} \left[\frac{s+j\omega - s+j\omega}{s^2 + \omega^2} \right] = \frac{\omega}{s^2 + \omega^2}$$

Properties of Laplace Transform

1) Linearity:

$$ax_1(t) + bx_2(t) \longleftrightarrow ax_1(s) + bx_2(s)$$

2) Time-shifting:

$$x(t-t_0) \xleftrightarrow{\Delta} X(s)e^{-st_0}$$

3) Frequency shifting:

$$e^{-at}x(t) \longleftrightarrow X(s+a)$$

4) Differentiation property:

$$\frac{dx(t)}{dt} \longleftrightarrow sX(s) - x(0)$$

$$\frac{d^2x(t)}{dt^2} \longleftrightarrow s^2X(s) - sx(0) - x'(0)$$

$$\frac{d^n x(t)}{dt^n} \longleftrightarrow s^n X(s) - \sum_{k=1}^n s^{n-k} x^{(k-1)}(0)$$

5) Integration property:

$$\int_0^t x(\tau) d\tau \longleftrightarrow \frac{X(s)}{s}$$

(3)

Initial value theorem

$$f(0^+) = \lim_{s \rightarrow \infty} sF(s)$$

Final value theorem

$$f(\infty) = \lim_{s \rightarrow 0} s(F(s))$$

→ 3 types of analysis

1- Transient 2- steady state 3- stability

$$\rightarrow \frac{d^2y(t)}{dt^2} + 12\frac{dy(t)}{dt} + 32y(t) = 32x(t)$$

if initial condition are zero

taking laplace transform

$$s^2 Y(s) + 12s Y(s) + 32 Y(s) = 32 X(s)$$

$$\frac{Y(s)}{X(s)} = \frac{32}{s^2 + 12s + 32}$$

Let say $x(t) = u(t)$, we want to find output

$$X(s) = u(s) = \frac{1}{s}$$

$$Y(s) = \frac{32}{s(s^2 + 12s + 32)} = \frac{32}{s(s+4)(s+8)}$$

$$= \frac{A}{s} + \frac{B}{s+4} + \frac{C}{s+8}$$

$$32 = A(s+4)(s+8) + Bs(s+8) + Cs(s+4)$$

$$s=0$$

$$32 = A(0+4)(0+8), \quad s=-8, \quad s=-4$$

$$A=1, \quad B=-2, \quad C=1$$

$$Y(s) = \frac{1}{s} - \frac{2}{s+4} + \frac{1}{s+8}$$

→ Now taking inverse Laplace transform

$$\begin{aligned} y(t) &= u(t) - 2e^{-4t}u(t) + e^{-8t}u(t) \\ &= u(t) [1 - 2e^{-4t} + e^{-8t}] \end{aligned}$$

$$\rightarrow F(s) = \frac{2}{(s+1)(s+2)^2}$$

$$= \frac{A}{s+1} + \frac{B}{s+2} + \frac{C}{(s+2)^2}$$

$$2 = A(s+2)^2 + B(s+1)(s+2)^2 + C(s+1)$$

$$s = -1, s = -2$$

$$2 = A(-1+2)^2 + B(-1+1)(-1+2)^2 + C(-1+1)$$

$$2 = A(1)^2 \Rightarrow \boxed{A=2}$$

$$2 = A(-2+2)^2 + B(-2+1)(-2+2)^2 + C(-2+1)$$

$$2 = C(-1) \Rightarrow \boxed{C=-2}$$

$$4A + 2B + C = 2$$

$$4(2) + 2B + (-2) = 2$$

$$\boxed{B=-2}$$

$$F(s) = \frac{2}{s+1} - \frac{2}{s+2} - \frac{2}{(s+2)^2}$$

$$f(t) = 2e^{-t}u(t) - 2e^{-2t}u(t) - 2e^{-2t}t u(t)$$

$$= 2u(t)[e^{-t} - e^{-2t} - te^{-2t}]$$

$$\rightarrow F(s) = \frac{3}{s(s^2+2s+5)}$$

$$= \frac{A}{s} + \frac{Bs+C}{s^2+2s+5}$$

$$3 = A(s^2+2s+5) + Bs+C(s)$$

$$s=0$$

$$\boxed{A=3}$$

Now,

$$A+B=0 \quad ; \quad 2A+C=0$$

$$B = \frac{-3}{5} ; C = \frac{-6}{5}$$

$$F(s) = \frac{3/5}{s} - \frac{3/5s + 6/5}{s^2 + 2s + 5}$$

$$= \frac{3/5}{s} - \frac{3}{5} \frac{s+2}{s^2 + 2s + 5} = \frac{3/5}{s} - \frac{3}{5} \frac{s+2}{s^2 + 2s + 1 + 4}$$

$$= \frac{3/5}{s} - \frac{3}{5} \frac{s+2}{(s+1)^2 + (2)^2}$$

$$= \frac{3/5}{s} - \frac{3}{5} \left[\frac{s+1}{(s+1)^2 + 2^2} + \frac{1}{(s+1)^2 + 2^2} \right]$$

$$F(s) = \frac{3/5}{s} - \frac{3}{5} \left[\frac{s+1}{(s+1)^2 + 2^2} + \frac{1}{2} \frac{2}{(s+1)^2 + 2^2} \right]$$

$$f(t) = \frac{3}{5} u(t) - \frac{3}{5} \left[e^{-t} \cos 2t + \frac{1}{2} e^{-t} \sin 2t \right] u(t)$$

$$= \left\{ \frac{3}{5} - \frac{3}{5} e^{-t} \left[\frac{\cos 2t + \frac{1}{2} \sin 2t}{\sqrt{1 + (\frac{1}{2})^2}} \right] u(t) \right\}$$

$$= 0.6 - 0.67 e^{-t} \left[\frac{1}{\sqrt{1 + (1/2)^2}} \cos 2t + \frac{1/2}{\sqrt{1 + (1/2)^2}} \sin 2t \right]$$

$$\therefore \cos(A-B) = \cos A \cos B + \sin A \sin B$$

$$= 0.6 u(t) - 0.67 e^{-t} \cos(at - \phi) u(t)$$

$$\phi = \tan^{-1} \frac{1}{2}$$

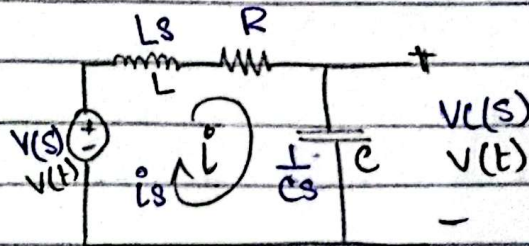
$$G(s) = \frac{2s+1}{s^2+2s+6}$$

$$(s^2+2s+6)Y(s) = (2s+1)X(s)$$

$$s^2 Y(s) + 2s Y(s) + 6Y(s) = 2sX(s) + X(s)$$

$$\frac{d^2 y(t)}{dt^2} + 2 \frac{dy(t)}{dt} + 6y(t) = 2 \frac{dx(t)}{dt} + x(t)$$

Transfer function of Electrical Networks:



$$\rightarrow i = C \frac{dv_c}{dt}$$

$$I(s) = Cs V(s)$$

$$\rightarrow V_L = L \frac{di}{dt}$$

Find $\frac{V_c(s)}{V(s)}$

$$V_L(s) = Ls I(s)$$

$$-V(t) + L \frac{di}{dt} + Ri + V_c(t) = 0$$

$$-V(t) + LC \frac{d^2 V_c(t)}{dt^2} + RC \frac{dV_c}{dt} + V_c(t) = 0$$

taking Laplace transform

$$LCs^2 V_c(s) + RCs V_c(s) + V_c(s) + V(s) = 0$$

$$\frac{V_c(s)}{V(s)} = \frac{-1}{LCs^2 + RCs + 1}$$

$$= \frac{-1/LC}{s^2 + \frac{R}{L}s + \frac{1}{LC}}$$

→ in terms of 's'

$$-V(s) + LsI(s) + RI(s) + I(s)\frac{1}{Cs} = 0$$

$$(Ls + R + \frac{1}{Cs})(I(s)) = V(s)$$

$$(Ls + R + \frac{1}{Cs})(Cs)(V_C(s)) = V(s)$$

$$(LCs^2 + RCs + 1)(V_C(s)) = V(s)$$

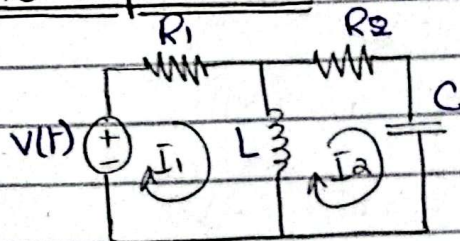
$$\frac{V_C(s)}{V(s)} = \frac{1}{LCs^2 + RCs + 1}$$

→ 3rd Method

Apply VDR:

$$V_C(s) = V(s) \frac{1/Cs}{1/Cs + R + Ls}$$

Multiple Loop Circuit



$$-V(s) + R_1 I_1(s) + Ls(I_1(s) - I_2(s)) = 0$$

$$R_2 I_2(s) + \frac{1}{Cs} I_2(s) + Ls(I_2(s) - I_1(s)) = 0$$

$$(Ls + R_1)I_1(s) - LsI_2(s) = V(s)$$

$$-LsI_1(s) + \left(\frac{1}{Cs} + (Ls + R_2)I_2(s)\right) = 0$$

$$\begin{bmatrix} Ls + R_1 & -Ls \\ -Ls & \frac{1}{Cs} + Ls + R_2 \end{bmatrix} \begin{bmatrix} I_1(s) \\ I_2(s) \end{bmatrix} = \begin{bmatrix} V(s) \\ 0 \end{bmatrix}$$

$$I_1(s) = \frac{\begin{vmatrix} V(s) & -Ls \\ 0 & R_2 + Ls + \frac{1}{Cs} \end{vmatrix}}{\Delta}$$

Δ

$$I_2(s) = \frac{\begin{vmatrix} Ls + R_1 & V(s) \\ -Ls & 0 \end{vmatrix}}{\Delta}$$

Δ

$$\Delta = \begin{vmatrix} Ls + R_1 & -Ls \\ -Ls & R_2 + Ls + \frac{1}{Cs} \end{vmatrix}$$

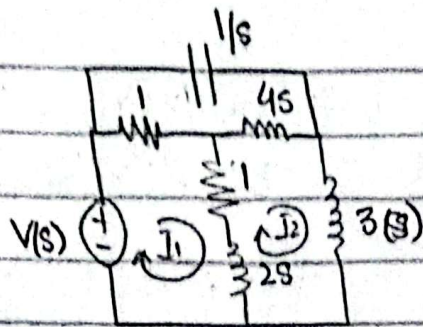
$$= (Ls + R_1)\left(R_2 + \frac{1}{Cs} + Ls\right) - L^2s^2$$

Shortcut Method:

$$\left[\begin{array}{c} \text{Sum of imp} \\ \text{around Mesh 1} \end{array} \right] I_1(s) - \left[\begin{array}{c} \text{sum of imp} \\ \text{common in} \\ \text{1 and 2} \end{array} \right] I_2(s) = \left[\begin{array}{c} \text{sum of} \\ \text{voltage in 1} \end{array} \right]$$

$$\left[\begin{array}{c} \text{sum of imp} \\ \text{around Mesh 2} \end{array} \right] I_2(s) - \left[\begin{array}{c} \text{sum of imp} \\ \text{common in} \\ \text{1 and 2} \end{array} \right] I_1(s) = \left[\begin{array}{c} \text{sum of} \\ \text{voltage in 2} \end{array} \right]$$

(4)



Find $\frac{I_1(s)}{V(s)}$

Mesh 1:

$$(2+2s)(I_1(s)) - (1+2s)(I_2(s)) - (1)(I_3(s)) = V_s \quad \text{--- (1)}$$

Mesh 2:

$$(1+9s)(I_2(s)) - (1+2s)(I_1(s)) - 4s(I_3(s)) = 0 \quad \text{--- (2)}$$

Mesh 3:

$$(1+4s+\frac{1}{s})(I_3(s)) - (1)(I_1(s)) - 4s(I_2(s)) = 0 \quad \text{--- (3)}$$

$$\begin{bmatrix} 2+2s & -(1+2s) & -1 \\ -(1+2s) & 9s+1 & -4s \\ -1 & -4s & \frac{1}{s}+4s+1 \end{bmatrix} \begin{bmatrix} I_1(s) \\ I_2(s) \\ I_3(s) \end{bmatrix} = \begin{bmatrix} V_s \\ 0 \\ 0 \end{bmatrix}$$

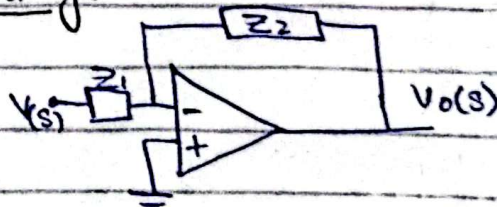
$$I_1(s) = \frac{\begin{vmatrix} V_s & -(1+2s) & -1 \\ 0 & (9s+1) & -4s \\ 0 & -4s & \frac{1}{s}+4s+1 \end{vmatrix}}{\Delta}$$

Δ

$$\Delta = \begin{vmatrix} 2+2s & -(1+2s) & -1 \\ -(1+2s) & 9s+1 & -4s \\ -1 & -4s & \frac{1}{s}+4s+1 \end{vmatrix}$$

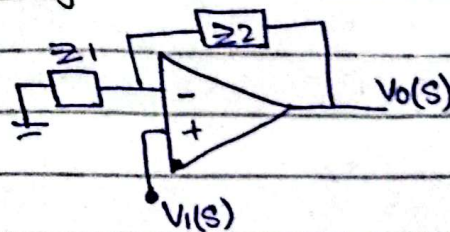
Transfer function of Active ccts / OP-AMP based ccts

Inverting:



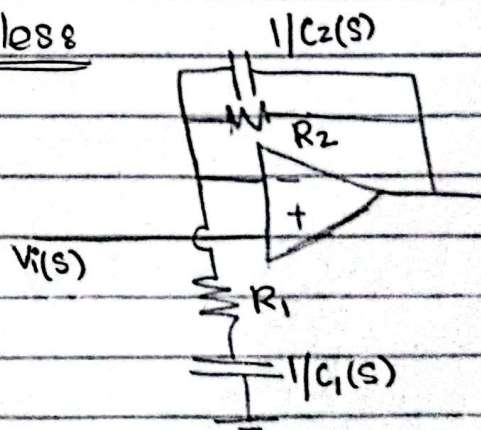
$$\frac{V_o(s)}{V_i(s)} = -\frac{Z_2}{Z_1}$$

Non-Inverting:



$$\frac{V_o(s)}{V_i(s)} = 1 + \frac{Z_2}{Z_1}$$

Examples:



$$Z_2 = R_2 \parallel \frac{1}{C_2(s)} = \frac{R_2}{R_2 C_2(s) s + 1}$$

$$= \frac{R_2}{1 + R_2 C_2(s)} \quad \text{--- ①}$$

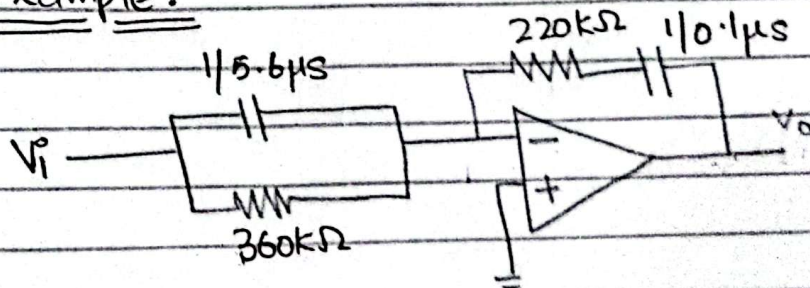
$$Z_1 = R_1 + \frac{1}{C_1(s)} = \frac{1 + R_1 C_1(s)}{C_1(s)} \quad \text{--- ②}$$

$$\begin{aligned}
 \frac{V_o(s)}{V_i(s)} &= 1 + \frac{R_2 / (1 + R_2 C_2(s))}{\frac{1 + R_1 C_1(s)}{C_1(s)}} \\
 &= 1 + \frac{R_2 C_1(s)}{(1 + R_2 C_2(s))(1 + R_1 C_1(s))} \\
 &= \frac{1 + (R_1 C_1(s)(1 + R_2 C_2(s)) + R_2 C_1 s)}{(1 + R_1 C_1 s)(1 + R_2 C_2 s)} \\
 &= \frac{R_1 C_1 R_2 C_2 s^2 + (R_1 C_1 + R_2 C_2 + R_2 C_1) s + 1}{R_1 C_1 R_2 C_2 s^2 + (R_1 C_1 + R_2 C_2) s + 1}
 \end{aligned}$$

∴ Poles = Power of s in denominator

∴ Zeros = Power of s in numerator

Example:



$$Z_2 = 220k + \frac{10^7}{s}, \quad Z_1 = \frac{360k}{1 + (360k)(5.6\mu)s}$$

$$\frac{V_o(s)}{V_i(s)} = \frac{220k + 10^7/s}{360k / (1 + (360k)(5.6\mu)s)}$$

$$= - \left[\frac{220ks + 10^7}{s} \times \left(\frac{(1 + 2.016)s}{360k} \right) \right]$$

$$= - \left(\frac{1.232s^2 + 56.61s + 27.77}{s} \right)$$

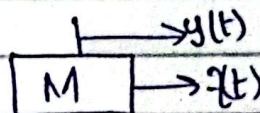
$$= - \left[1.232s + \frac{27.77}{s} + 56.61 \right]$$

Transfer Function of Mechanical Systems

→ Translational Mechanical System → Mass, Spring, Damper
 ↳ Force, Displacement

Mass:

→ stores kinetic energy
 → rigid bodies

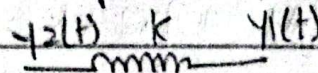
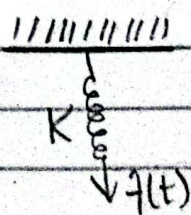


$$f(t) \propto a$$

$$f(t) = Ma \Rightarrow \boxed{f(t) = M \frac{d^2 y(t)}{dt^2}}$$

2- Spring:

→ stores potential energy



$$f(t) = k(y_1(t) - y_2(t))$$

$$f(t) \propto y(t)$$

$$\boxed{f(t) = ky(t)}$$

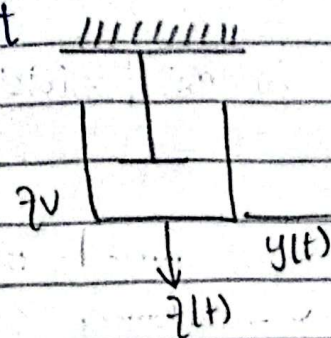
3- Damper:

→ Frictional element

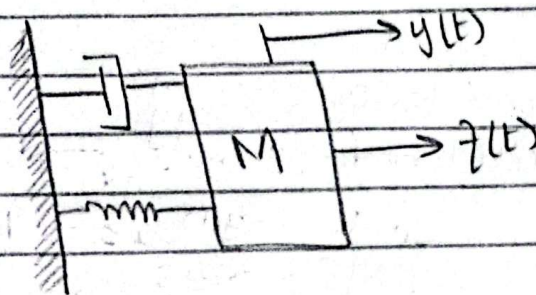
$$\begin{aligned} \tau(t) &\propto v(t) \\ \tau(t) &\propto \frac{dy(t)}{dt} \end{aligned}$$

$$\tau(t) = \tau_v \frac{dy(t)}{dt}$$

→ τ_v = viscous friction coefficient



Ideal Spring Mass Damper System



$\begin{matrix} M \frac{d^2 y}{dt^2} \leftarrow \\ \tau_v \frac{dy}{dt} \leftarrow \\ k y \leftarrow \end{matrix} \begin{matrix} \text{FBD} \\ \text{at } y_1 \end{matrix} \rightarrow \tau(t) \quad \tau(t) = M \frac{d^2 y}{dt^2} + \tau_v \frac{dy}{dt} + k y(t)$

Taking Laplace transform

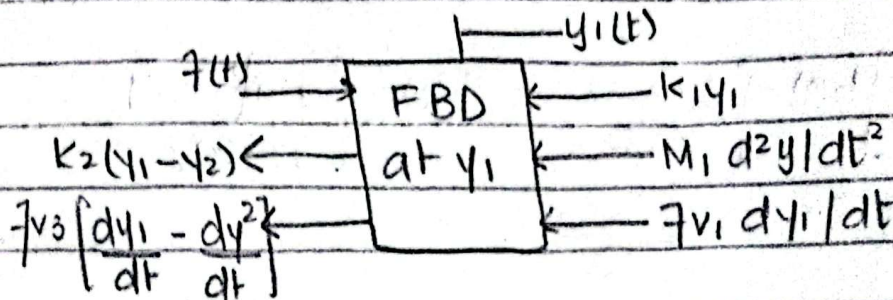
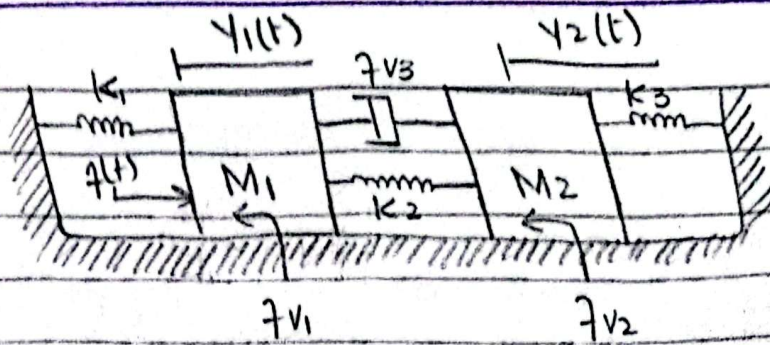
$$F(s) = Ms^2 Y(s) + \tau_v s Y(s) + k Y(s)$$

$$Y(s) [Ms^2 + \tau_v s + k] = F(s)$$

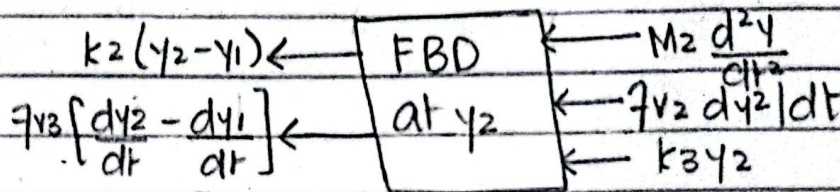
$$\frac{Y(s)}{F(s)} = \frac{1}{Ms^2 + \tau_v s + k}$$

$$M=3, \tau_v=2, k=1$$

$$\frac{Y(s)}{F(s)} = \frac{1}{3s^2 + 2s + 1}$$



$$f(t) = M_1 \frac{d^2 y_1}{dt^2} + \gamma_{v1} \frac{dy_1}{dt} + \gamma_{v3} \left[\frac{dy_1}{dt} - \frac{dy_2}{dt} \right] + k_1 y_1 + k_2 (y_1 - y_2)$$



$$M_2 \frac{d^2 y_2}{dt^2} + \gamma_{v2} \frac{dy_2}{dt} + k_3 y_2 + k_2 (y_2 - y_1) + \gamma_{v3} \left[\frac{dy_2}{dt} - \frac{dy_1}{dt} \right] = 0$$

$$\Rightarrow [M_1 s^2 (\gamma_{v1} + \gamma_{v3}) s + (k_1 + k_2)] Y_1(s) - (\gamma_{v3} s + k_2) Y_2(s) = F(s)$$

$$\Rightarrow -(\gamma_{v3} s + k_2) Y_1(s) + [M_2 s^2 + (\gamma_{v2} + \gamma_{v3}) s + k_2 + k_3] Y_2(s) = 0$$

CRAMER'S RULE:

$$\begin{bmatrix} M_1 s^2 + (\gamma_{v1} + \gamma_{v3}) s + k_1 + k_2 & -\gamma_{v3} s + k_2 \\ -(\gamma_{v3} s + k_2) & M_2 s^2 + (\gamma_{v2} + \gamma_{v3}) s + k_2 + k_3 \end{bmatrix}$$

$$\begin{bmatrix} Y_1(s) \\ Y_2(s) \end{bmatrix} = \begin{bmatrix} F(s) \\ 0 \end{bmatrix}$$

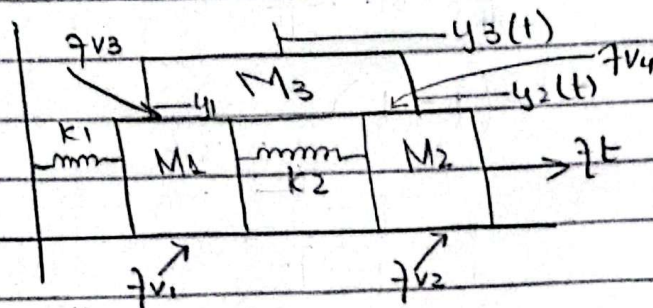
* Technique:

$$\left[\text{sum of imp at motion 1} \right] Y_1(s) - \left[\text{sum of imp common in 1 and 2} \right] Y_2(s) = \left[\text{sum of applied forces at } Y_1 \right]$$

$$\left[\text{sum of imp at motion 2} \right] Y_2(s) - \left[\text{sum of imp common} \right] Y_1(s) = \left[\text{sum of applied forces at } Y_2 \right]$$

(6)

Example:



Equation by using technique.

For M1:

$$(M_1 s^2 + (f_{v1} + f_{v3})s + k_1 + k_2) Y_1(s) - k_2 Y_2(s) - (f_{v3} s) Y_3(s) = 0$$

For M2:

$$(M_2 s^2 + (f_{v2} + f_{v4})s + k_2) Y_2(s) - k_2 Y_1(s) - (f_{v4} s) Y_3(s) = F(s)$$

For M3:

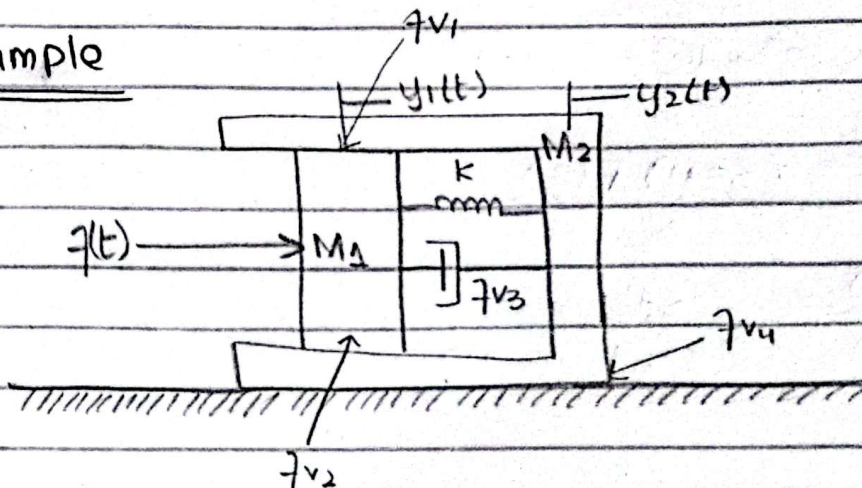
$$(M_3 s^2 + (f_{v3} + f_{v4})s) Y_3(s) - (f_{v3} s) Y_1(s) - (f_{v4} s) Y_2(s) = 0$$

Equation using Cramer's rule:

$$\begin{bmatrix} M_1 s^2 + (\gamma_{v1} + \gamma_{v3})s + k_1 + k_2 & -k_2 & -\gamma_{v3}s \\ M_2 s^2 + (\gamma_{v2} + \gamma_{v4})s + k_2 & -\gamma_{v4}s & M_3 s^2 + (\gamma_{v3} - \gamma_{v4})s \end{bmatrix} \begin{bmatrix} y_1(s) \\ y_2(s) \\ y_3(s) \end{bmatrix} = \begin{bmatrix} 0 \\ F(s) \\ 0 \end{bmatrix}$$

Remember!
Matrix mein M, k, γ ki value put kar niha

Example



Take $\gamma_{v1} = \gamma_{v2} = \gamma_{v3} = \gamma_{v4} = 1 \text{ N s/m}$

$M_1 = M_2 = 1 \text{ kg}$

$k = 1 \text{ N/m}$

Find $\frac{y_2(s)}{F(s)}$

For M1:

$$(M_1 s^2) y_1(s) - (\gamma_{v1} + \gamma_{v2} + \gamma_{v3} + k) y_2(s) = F(s)$$

$$+ (\gamma_{v1} + \gamma_{v2} + \gamma_{v3})s + k$$

$$\rightarrow (M_1 s^2 + (\gamma_{v1} + \gamma_{v2} + \gamma_{v3})s + k) y_1(s)$$

For M_2 :

$$(M_2 s^2 + (\gamma v_1 + \gamma v_2 + \gamma v_3 + \gamma v) s + k) - [(\gamma v_1 + \gamma v_2 + \gamma v_3) s + k] y_1(s) = 0$$

Cramer's Rule

$$\begin{bmatrix} s^2 + 3s + 1 & -(3s + 1) \\ -(3s + 1) & s^2 + 4s + 1 \end{bmatrix} \begin{bmatrix} y_1(s) \\ y_2(s) \end{bmatrix} = \begin{bmatrix} F(s) \\ 0 \end{bmatrix}$$

$$y_2(s) = \frac{\begin{vmatrix} s^2 + 3s + 1 & F(s) \\ -(3s + 1) & 0 \end{vmatrix}}{\Delta}$$

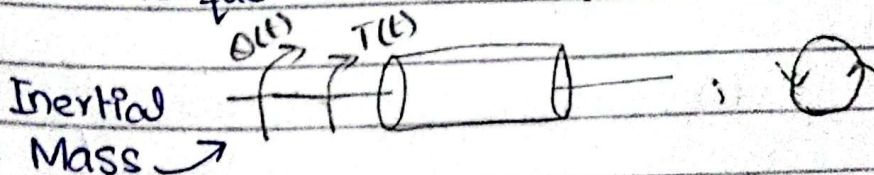
$$\Delta = s(s^3 + 7s^2 + 5s + 1)$$

$$y_2(s) = \frac{(3s + 1) F(s)}{\Delta}$$

$$y_2(s) = \frac{3s + 1}{s(s^3 + 7s^2 + 5s + 1)} F(s)$$

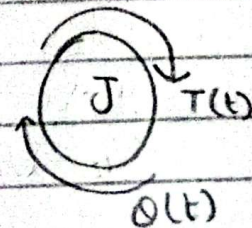
Rotational Mechanical System

→ Torque $T(t)$ → Angular displacement $\theta(t)$

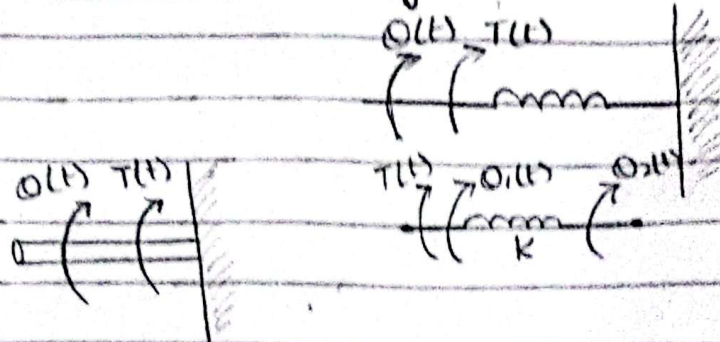


$$T(t) \propto \frac{d^2 \theta(t)}{dt^2} ; T(t) = J \frac{d^2 \theta(t)}{dt^2}$$

J = Moment of Inertia



→ Torsional Spring:



$$T(t) \propto \theta(t)$$

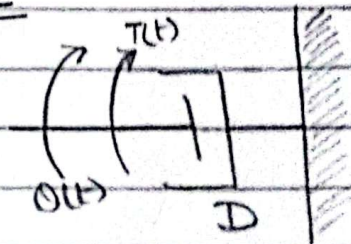
$$T(t) = k\theta(t)$$

$$T(t) = k(\theta_1(t) - \theta_2(t))$$

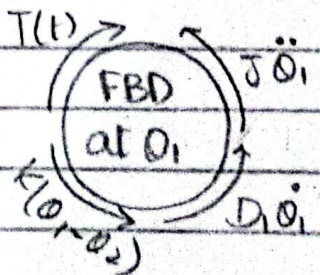
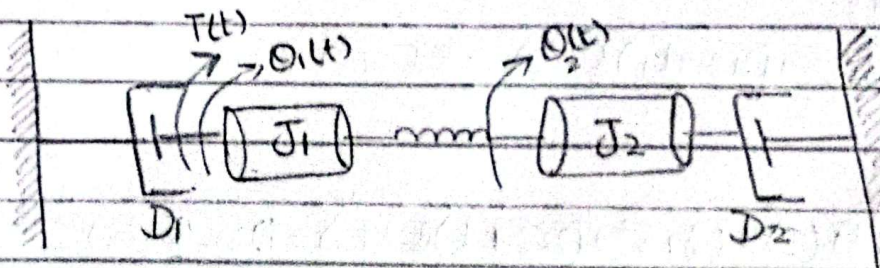
→ Torsional damper:

$$T(t) \propto \frac{d\theta(t)}{dt}$$

$$T(t) = D \frac{d\theta(t)}{dt}$$

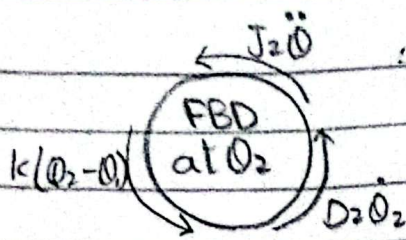


Example



$$J_1 \frac{d^2 \theta_1}{dt^2} + D_1 \frac{d\theta_1}{dt} + k(\theta_1 - \theta_2) = T(t)$$

$$\hookrightarrow (J_1 s^2 + D_1 s + k) \theta_1(s) - k \theta_2(s) = T(s)$$



$$J_2 \frac{d^2 \theta_2}{dt^2} + D_2 \frac{d \theta_2}{dt} + k(\theta_2 - \theta_1) = 0$$

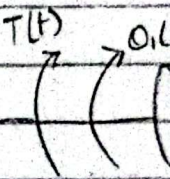
$$-k \theta_1(s) + (J_2 s^2 + D_2 s + k) \theta_2(s) = 0$$

Cramer's

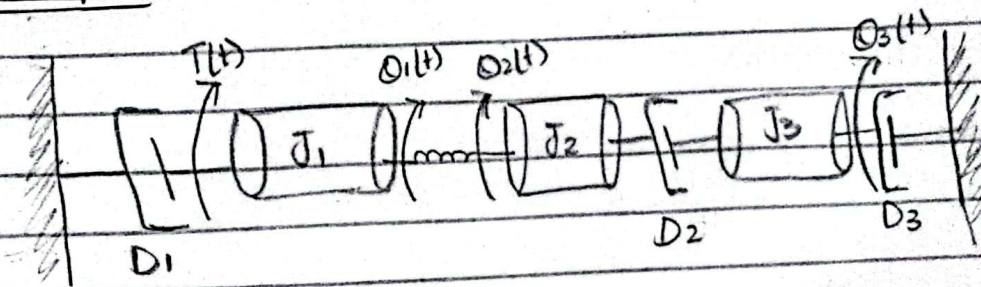
$$\begin{bmatrix} J_1 s^2 + D_1 s + k & -k \\ -k & J_2 s^2 + D_2 s + k \end{bmatrix} \begin{bmatrix} \theta_1 \\ \theta_2 \end{bmatrix} = \begin{bmatrix} T(s) \\ 0 \end{bmatrix}$$

Cramer's Rule:

$$\begin{bmatrix} J_1 s^2 + D_1 s + k & -k \\ -k & J_2 s^2 + D_2 s + k \end{bmatrix} \begin{bmatrix} \theta_1 \\ \theta_2 \end{bmatrix} = \begin{bmatrix} T(s) \\ 0 \end{bmatrix}$$



Example:



For θ_1 :

$$(J_1 s^2 + D_1 s + k_1) \theta_1(s) - k \theta_2(s) = T(s)$$

For θ_2 :

$$-k \theta_1(s) + (J_2 s^2 + D_2 s + k) \theta_2(s) - D_2 s \theta_3(s) = 0$$

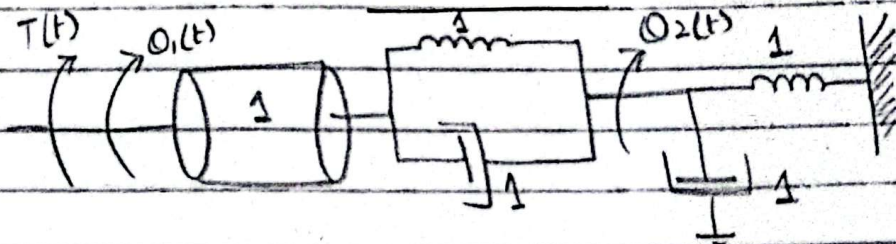
$\Delta =$

For θ_3 :

$$-D_2 s \theta_2(s) + (J_3 s^2 + (D_2 + D_3) s) \theta_3(s) = 0$$

Cramers' Rule :

$$\begin{bmatrix} J_1 s^2 + D_1(s) + k & -k & 0 \\ -k & J_2 s + D_2 s + k & -D_2 s \\ 0 & -D_2 s & J_3 \theta_2 + (D_2 + D_3)s \end{bmatrix} \begin{bmatrix} \theta_1(s) \\ \theta_2(s) \\ \theta_3(s) \end{bmatrix} = \begin{bmatrix} T(s) \\ 0 \\ 0 \end{bmatrix}$$



Find $\frac{\theta_2(s)}{T(s)}$

$$\begin{aligned} (s^2 + s + 1)\theta_1(s) - (s + 1)\theta_2(s) &= T(s) \\ -(s + 1)\theta_1(s) + (2s + 2)\theta_2(s) &= 0 \end{aligned}$$

Cramer's Rule:

$$\begin{bmatrix} s^2 + s + 1 & -(s + 1) \\ -(s + 1) & 2s + 2 \end{bmatrix} \begin{bmatrix} \theta_1 \\ \theta_2 \end{bmatrix} = \begin{bmatrix} T(s) \\ 0 \end{bmatrix}$$

$$\theta_2(s) = \frac{\begin{vmatrix} s^2 + s + 1 & T(s) \\ -(s + 1) & 0 \end{vmatrix}}{\Delta}$$

$$\Delta = \begin{vmatrix} s^2 + s + 1 & -(s + 1) \\ -(s + 1) & 2s + 2 \end{vmatrix}$$